Survival MF Derivations

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Contents

1 Introduction

The following document presents derivations for matrix factorization & survival expressions for supervised matrix factorization. Content includes an expression of the overall objective function to minimize as well as the procedure for updating the prior distributions for L, F, β, τ. These updates are solved via the EBNM problem illustrated in the EBMF paper. This document incorporates corrections to the notation and derivations from the 3/6/26 version (V3: Option A — R_{ij}^{-k} and z_i^{-k} are defined using raw components; posterior-mean bars $\overline{R}_{ij}^{-k}$ and $\overline{z}_i^{-k}$ are applied only in the final coefficient expressions).

1A Models

The following models are considered as part of this supervised matrix factorization task:

- Genomics Data: $Y_{n\times p} = L_{n\times K}F'_{K\times p} + E_{n\times p}$, where $E_{ij} \overset{\text{iid}}{\sim} N\left(0, \tau_j^{-1}\right)$
- Survival Data: $h(t_i) = h_0(t_i)\exp\left(\sum_k l_{ik}\beta_k\right)$

1B Update Procedure

The general procedure for parameter (prior & posterior distribution) updates is as follows.

For b=1,$\cdots$, B

1. Compute survival working quantities $\hat{\eta}_i$, u_i, W_{ii}, z_i from the current posterior means.
2. Iterate over $k = 1, \cdots K$ for each of the following.
3. update $q(l_k), g(l_k)$ with other parameters fixed (different from EBMF problem: combines genomics & survival terms)
4. update $q(f_k), g(f_k)$ with other parameters fixed (identical to EBMF problem)
5. update $q(\beta_k), g(\beta_k)$ with other parameters fixed (survival term only; EBNM under mean-field independence)
6. update τ_j (identical to EBMF problem)

2 Objective Function

The objective function to minimize that considers priors (g) & posteriors (q) for all parameters is

$$F\left(\underbrace{q_l, q_f, q_\beta}_{posteriors}, \underbrace{g_l, g_f, g_\beta}_{priors}, \tau\right) = E_{q_l, q_f, q_\beta}\left[\log P\left(\underbrace{Y, t, \delta}_{observed} | \underbrace{L, F, \beta, \tau}_{unknown}\right)\right] + \sum_k E_{q_{l_k}}\left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)}\right] + \sum_k E_{q_{f_k}}\left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)}\right] + \sum_k E_{q_\beta}\left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)}\right] \quad (1)$$

and the full data likelihood is expressed as

$$\log P\left(\underbrace{Y}_{Genomics}, \underbrace{t, \delta}_{Survival} | L, F, \beta, \tau\right) = \underbrace{\log P\left(Y|L, F, \tau^{-1}\right)}_{Genomics} + \underbrace{\log P\left(t, \delta|L, \beta\right)}_{Survival} \quad (2)$$

3 Taylor Series Approximation of Survival Likelihood

Now we consider a 2nd-Order Taylor Series Expansion of the survival data likelihood for the sake of finding a tractable solution.

First, note that the 2nd-Order approximation takes the form

$$f(x) = f(x_0) + \underbrace{f'(x_0)}_{gradient/score} (x - x_0) + \frac{1}{2} \underbrace{f''(x_0)}_{hessian/info} (x - x_0)^2 \tag{3}$$

and that the linear predictor for the survival data has the form

$$\eta_i = l_i'\beta = \sum_k l_{ik}\beta_k \tag{4}$$

Let $\hat{\boldsymbol{\eta}}$ denote the current expansion point, and define the score u_i and negative diagonal Hessian W_{ii} as

$$u_i = \left.\frac{\partial \ell}{\partial \eta_i}\right|_{\hat{\eta}_i}, \qquad W_{ii} = -\left.\frac{\partial^2 \ell}{\partial \eta_i^2}\right|_{\hat{\eta}_i} \geq 0 \tag{5}$$

Then the Approximation is as follows

$$\begin{aligned}
&\log P(t, \delta | L, \beta) = \ell(\eta) && (6)\\
&\approx \ell(\hat{\eta}) + (\eta - \hat{\eta})^\top \underbrace{\nabla \ell(\hat{\eta})}_{\mathbf{U}} + \frac{1}{2} (\eta - \hat{\eta})^\top \underbrace{\nabla^2 \ell(\hat{\eta})}_{\mathbf{H}=-\mathbf{W}} (\eta - \hat{\eta}) && (7)\\
&\approx \ell(\hat{\eta}) + \sum_i (\eta_i - \hat{\eta}_i)\, \mathbf{u_i} + \frac{1}{2}\sum_i (\eta_i - \hat{\eta}_i)^2\, \mathbf{H}_{ii} \quad \left(\text{quadratic form: } \sum_i a x_i^2 + b x_i + c \to a = ?, b = ?\right) && (8)\\
&= \sum_i \left[\mathbf{u}_i (\eta_i - \hat{\eta}_i) - \frac{1}{2}\mathbf{W}_{ii}\left(\eta_i^2 - 2\eta_i\hat{\eta}_i + \hat{\eta}^2\right)\right] && (9)\\
&= \sum_i \left[\mathbf{u}_i\eta_i - \mathbf{u}_i\hat{\eta}_i - \frac{1}{2}\mathbf{W}_{ii}\eta_i^2 + \mathbf{W}_{ii}\eta_i\hat{\eta}_i - \frac{1}{2}\mathbf{W}_{ii}\hat{\eta}^2\right] \quad (\text{keep only terms with } \eta_i) && (10)\\
&\text{where} \quad \sum_i \left(a_i\eta_i^2 + b_i\eta_i\right) = \sum_i a_i \left(\eta_i + \frac{1}{2}\frac{b_i}{a_i}\right)^2 && (11)\\
&= \sum_i \left[\left(\underbrace{-\frac{1}{2}\mathbf{W}_{ii}}_{A}\right)\eta_i^2 + \left(\underbrace{\mathbf{u}_i + \mathbf{W}_{ii}\hat{\eta}_i}_{B}\right)\eta_i\right] && (12)\\
&= \sum_i \left(-\frac{1}{2}\mathbf{W}_{ii}(\eta_i - z_i)^2\right) && (13)\\
&\text{where} \quad B = \mathbf{W}_{ii} z_i = \mathbf{u}_i + \mathbf{W}_{ii}\hat{\eta}_i \Rightarrow \boxed{z_i = \frac{\mathbf{u}_i}{\mathbf{W}_{ii}} + \hat{\eta}_i} && (14)\\
&= \sum_i \left[-\frac{1}{2}\mathbf{W}_{ii}\left(\eta_i - \left(\underbrace{\frac{\mathbf{u}_i}{\mathbf{W}_{ii}} + \hat{\eta}_i}_{z_i}\right)\right)^2\right] && (15)
\end{aligned}$$

4 Update for q_l

For the updates of q_l & g_l, we consider terms from F that pertain to l. Therefore, we consider

$$F(q_l, g_l) = E_{q_l,q_f,q_\beta}\left[\log P(Y,t,\delta|L,F,\beta,\tau)\right] + \sum_k E_{q_{l_k}}\left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)}\right] \tag{16}$$

$$= \underbrace{E_{q_l,q_f,q_\beta}\left[\log P\left(Y|L,F,\tau^{-1}\right)\right]}_{\text{Equation 1}} + \underbrace{E_{q_l,q_f,q_\beta}\left[\log P(t,\delta|L,\beta)\right]}_{\text{Equation 2}} + \underbrace{\sum_k E_{q_{l_k}}\left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)}\right]}_{\text{Keep from above}} \tag{17}$$

$$= \sum_k \left[\text{Terms from } 1(l_k) + \text{Terms from } 2(l_k) + E_{q_{l_k}}\left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)}\right]\right] \tag{18}$$

Now, we need expressions for the terms of equations 1 & 2 that are dependent on l_k and/or l_k^2. To aid this process, we first consider the residuals for y_{ij}, denoted as R_{ij}.

Correction Note

Items 1 & 2 (Option A): The previous document defined R_{ij}^k and z_i^k using inconsistent superscript notation (k vs. $-k$). In this version (Option A), the partial residuals R_{ij}^{-k} and z_i^{-k} are *defined* using raw (unbarred) component values. When expectations with respect to q are taken in the EBNM coefficient expressions, this naturally produces the posterior-mean quantities $\overline{R}_{ij}^{-k}$ and $\overline{z}_i^{-k}$, which appear with a bar only in those final coefficient expressions. This is the minimal correction to the 3/6/26 structure: the definitions are unchanged in form, and bars are added only where required by taking expectations. Note also that z_i^{-k} involves $\boldsymbol{\beta}$, not F — this corrects an inconsistency in the prior document where $f_{jk'}$ appeared in a per-sample survival quantity.

The k-th genomics partial residual, evaluated at posterior means, is:

$$R_{ij}^{-k} = y_{ij} - \sum_{k' \neq k} l_{ik'} f_{jk'} \tag{19}$$

where the k-th residual excludes the contribution of factor k, using raw (unbarred) component values. When the expectation with respect to q is taken in the EBNM coefficient expressions below, the identity $E_q\left[R_{ij}^{-k}\right] = y_{ij} - \sum_{k' \neq k} \overline{l_{ik'}}\,\overline{f_{jk'}} =: \overline{R}_{ij}^{-k}$ produces the posterior-mean partial residual $\overline{R}_{ij}^{-k}$. Similarly, the k-th partial working response for the survival term is

$$z_i^{-k} = z_i - \sum_{k' \neq k} l_{ik'} \beta_{k'} \tag{20}$$

whose posterior mean is $\overline{z}_i^{-k} = z_i - \sum_{k' \neq k} \overline{l_{ik'}}\,\overline{\beta_{k'}}$.

Note: the EBMF paper defines R_{ij}^k as the posterior first moment of the residual (using barred components); in Option A we define R_{ij}^{-k} with raw components, and the barred version $\overline{R}_{ij}^{-k}$ arises naturally when expectations are taken..

We can now use these results to simplify the procedure of identifying the necessary terms of equations 1 & 2 to specify coefficients A and B attached to l_{ik} and l_{ik}^2.

First consider the log-likelihood term

$$E_{q_l,q_f,q_\beta}\left[\log P\left(Y|L,F,\tau^{-1}\right)\right] = E_{q_l,q_f,q_\beta}\left[\log\left(\prod_{ij} \frac{\tau_j}{\sqrt{2\pi}} \exp\left[-\frac{\tau_j}{2}\left(y_{ij} - \underbrace{\sum_k l_{ik} f_{jk}}_{\hat{Y}}\right)^2\right]\right)\right] \tag{21}$$

where the observed values are $y_{ij} = \sum_k l_{ik} f_{jk} + \varepsilon_{ij}$ and $\varepsilon \sim N\left(0, \sigma^2 = \tau^{-1}\right)$

Equation 1 is then expressed as

$$E_{q_l,q_f,q_\beta}\left[\log P\left(Y|L,F,\tau^{-1}\right)\right] = E_{q_l,q_f,q_\beta}\left[\sum_{ij}\left[-\frac{1}{2}\tau_j\left(y_{ij}-\sum_k l_{ik}f_{jk}\right)^2\right]\right] \tag{22}$$

$$= E_{q_l,q_f,q_\beta}\left[\sum_{ij}\left[-\frac{1}{2}\tau_j\left(R_{ij}^{-k}-l_{ik}f_{jk}\right)^2\right]\right] \tag{23}$$

$$= E_{q_l,q_f,q_\beta}\left[\sum_{ij}\left[-\frac{1}{2}\tau_j\left(\cancel{R_{ij}^{-k^2}}-2R_{ij}^{-k}\underline{l_{ik}}f_{jk}+\underline{l_{ik}^2}f_{jk}^2\right)\right]\right] \tag{24}$$

Similarly, equation 2 is expressed as

$$E_{q_l,q_f,q_\beta}\left[\log P\left(t,\delta|L,\beta\right)\right] = E_{q_l,q_f,q_\beta}\left[\sum_i\left[-\frac{1}{2}\mathbf{W}_{ii}\left(\sum_k \underbrace{l_{ik}\beta_k}_{\eta_i}-\underbrace{\left(\frac{\mathbf{u}_i}{\mathbf{W}_{ii}}+\hat{\eta}_i\right)}_{z_i}\right)^2\right]\right] \tag{25}$$

$$= E_{q_l,q_f,q_\beta}\left[\sum_i\left[-\frac{1}{2}\mathbf{W}_{ii}\left(z_i-\sum_k l_{ik}\beta_k\right)^2\right]\right] \tag{26}$$

$$= E_{q_l,q_f,q_\beta}\left[\sum_i\left[-\frac{1}{2}\mathbf{W}_{ii}\left(z_i^{-k}-l_{ik}\beta_k\right)^2\right]\right] \tag{27}$$

$$= E_{q_l,q_f,q_\beta}\left[\sum_i\left[-\frac{1}{2}\mathbf{W}_{ii}\left(\cancel{z_i^{-k^2}}-2z_i^{-k}\underline{l_{ik}}\beta_k+\underline{l_{ik}^2}\beta_k^2\right)\right]\right] \tag{28}$$

Now, we want to sum these two results and identify the coefficients A and B for the terms l_{ik} and l_{ik}^2 that satisfy the expression $A\cdot l_{ik}^2+B\cdot l_{ik}$. It follows that (considering the result of equation 18)

$$\begin{aligned} F(q_l,g_l) = \sum_k\Bigg[E_{q_l,q_f,q_\beta}\Bigg[&\sum_{ij}\left[-\frac{1}{2}\tau_j\left(\cancel{R_{ij}^{-k^2}}-2R_{ij}^{-k}l_{ik}f_{jk}+l_{ik}^2f_{jk}^2\right)\right] \\ &+\sum_i\left[-\frac{1}{2}\mathbf{W}_{ii}\left(\cancel{z_i^{-k^2}}-2z_i^{-k}l_{ik}\beta_k+l_{ik}^2\beta_k^2\right)\right]\Bigg] \\ &+E_{q_{l_k}}\left[\log\frac{g_{l_k}(l_k)}{q_{l_k}(l_k)}\right]\Bigg] \end{aligned} \tag{29}$$

Then for the genomics data (equation 1), the coefficients are

- l_{ik} coefficient (B): $\tau_j\overline{R_{ij}^{-k}}E\left[f_{jk}\right]=\tau_j\overline{R_{ij}^{-k}}\,\overline{f_{jk}}$
- l_{ik}^2 coefficient (A): $\tau_j E\left[f_{jk}^2\right]=\tau_j\overline{f_{jk}^2}$

and for the survival data (equation 2), the coefficients are

- l_{ik} coefficient (B): $\mathbf{W}_{ii}\overline{z_i^{-k}}E\left[\beta_k\right]=\mathbf{W}_{ii}\overline{z_i^{-k}}\,\overline{\beta_k}$
- l_{ik}^2 coefficient (A): $\mathbf{W}_{ii}E\left[\beta_k^2\right]=\mathbf{W}_{ii}\overline{\beta_k^2}$

such that the combined coefficients are

$$\mathbf{A_{ik}} = \sum_j\tau_j\overline{f_{jk}^2}+\mathbf{W}_{ii}\overline{\beta_k^2} \tag{30}$$

$$\mathbf{B_{ik}} = \sum_j\tau_j\overline{R_{ij}^{-k}}\,\overline{f_{jk}}+\mathbf{W}_{ii}\overline{z_i^{-k}}\,\overline{\beta_k} \tag{31}$$

Then, for the EBNM problem with $EBNM(x, s)$ where $x_i = \frac{B_{ik}}{A_{ik}}$ and $s_i^2 = \frac{1}{A_{ik}}$, we have that

$$x_i = \frac{\sum_j \tau_j \overline{R_{ij}^{-k}}\,\overline{f_{jk}} + \mathbf{W}_{ii}\overline{z_i^{-k}}\,\overline{\beta_k}}{\sum_j \tau_j \overline{f_{jk}^2} + \mathbf{W}_{ii}\overline{\beta_k^2}} \tag{32}$$

$$s_i^2 = \frac{1}{\sum_j \tau_j \overline{f_{jk}^2} + \mathbf{W}_{ii}\overline{\beta_k^2}} \tag{33}$$

and the EBNM problem can be solved via the EBNM R package.

5 Update for q_f

Updates for the prior & posterior of F follow the process as expressed in the EBMF paper. We want to find an update for the variational posterior $q_{f_k}(f_k)$ and prior g_{f_k} that maximize the objective function $F(\cdot)$ with all other parameters held fixed. To write out the update for q_f, note that F is only present in the term in the objective function for the genomics data (and isn't in the survival data term) such that we consider

$$F\left(q_{f_k}\right) \approx E_{q_l,q_f,q_\beta}\left[\log P\left(Y|L,F,\tau\right)\right] + E_{q_l,q_f,q_\beta}\left[\log g_{f_k}\left(f_k\right) - \log q_{f_k}\left(f_k\right)\right]. \tag{34}$$

The genomics data model is

$$Y_{ij} \sim N\left(\sum_k l_{ik} f_{jk}, \tau_j^{-1}\right). \tag{35}$$

Then expand the log-likelihood to be

$$E_{q_l,q_f,q_\beta}\left[\log P\left(Y|L,F,\tau\right)\right] = \sum_{ij} E_{q_l,q_f,q_\beta}\left[-\frac{1}{2}\tau_j\left(y_{ij} - \sum_k l_{ik} f_{jk}\right)^2\right] \tag{36}$$

Correction Note

Item 2 (Option A): The partial residual for the F update reuses the same raw-component definition as for the L update (Eq. 19), with the consistent $-k$ superscript notation. As in Section 4, taking the expectation with respect to q replaces the symbolic R_{ij}^{-k} with its posterior mean $\overline{R_{ij}^{-k}}$ in the final coefficient $\mathbf{B_{jk}}$ below.

and isolate the k-th factor by defining the residual excluding factor k (using raw components, as in Eq. 19) as

$$R_{ij}^{-k} = y_{ij} - \sum_{k' \neq k} l_{ik'} f_{jk'} \tag{37}$$

such that

$$y_{ij} = \left(\underbrace{y_{ij} - \sum_{k' \neq k} l_{ik'} f_{jk'}}_{R_{ij}^{-k}}\right) - l_{ik} f_{jk} + l_{ik} f_{jk} = R_{ij}^{-k} + l_{ik} f_{jk}. \tag{38}$$

Then substitute this result back into the likelihood expression to get

$$E_{q_l,q_f,q_\beta}\left[\log P\left(Y|L,F,\tau\right)\right] = \sum_{ij} E_{q_l,q_f,q_\beta}\left[-\frac{1}{2}\tau_j\left(R_{ij}^{-k} - l_{ik} f_{jk}\right)^2\right]. \tag{39}$$

We can then expand the squared term to isolate f_{jk} in the likelihood

$$E_{q_l,q_f,q_\beta}\left[\log P\left(Y|L,F,\tau\right)\right] = \sum_{ij} E_{q_l,q_f,q_\beta}\left[-\frac{1}{2}\tau_j\left(\left(R_{ij}^{-k}\right)^2 - 2R_{ij}^{-k} l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2\right)\right]. \tag{40}$$

and consider the expectation $E_q[\cdot]$ with respect to L, treating the residual R_{ij}^{-k} as fixed, so that

$$E_{q_l,q_f,q_\beta}\left[\log P\left(Y|L,F,\tau\right)\right] = \sum_{ij} E_{q_l,q_f,q_\beta}\left[-\frac{1}{2}\tau_j\left[\cancel{\left(R_{ij}^{-k}\right)^2} - 2R_{ij}^{-k} l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2\right]\right]. \tag{41}$$

Now, group terms to identify coefficients for f and f^2 to get the quadratic form $-\frac{1}{2}Af^2 + Bf$ (summing over i for each j to update f_k)

$$F\left(q_{f_k}\right) \approx E_{q_l,q_f,q_\beta}\left[\sum_j\left[-\frac{1}{2}f_{jk}^2\left(\underbrace{\sum_i \tau_j \overline{l_{ik}^2}}_{A_{jk}}\right) + f_{jk}\left(\underbrace{\sum_i \tau_j \overline{R_{ij}^{-k}}\,\overline{l_{ik}}}_{B_{jk}}\right)\right]\right] + E_{q_{f_k}}\left[\log\frac{g_{f_k}(f_k)}{q_{f_k}(f_k)}\right]. \tag{42}$$

Therefore, we have

$$\mathbf{A_{jk}} = \sum_i \tau_j \overline{l_{ik}^2} \tag{43}$$

$$\mathbf{B_{jk}} = \sum_i \tau_j \overline{R_{ij}^{-k}}\,\overline{l_{ik}} \tag{44}$$

which yield the following inputs for the EBNM problem

$$x_j = \frac{\mathbf{B_{jk}}}{\mathbf{A_{jk}}} = \frac{\sum_i \tau_j \overline{R_{ij}^{-k}}\,\overline{l_{ik}}}{\sum_i \tau_j \overline{l_{ik}^2}} \tag{45}$$

$$s_j^2 = \frac{1}{\mathbf{A_{jk}}} = \frac{1}{\sum_i \tau_j \overline{l_{ik}^2}} \tag{46}$$

6 Update for q_β

Correction Note

Item 3: The previous document contained two approaches for updating β: a weighted least-squares (WLS) formulation and an EBNM formulation (formerly Sec. 6A). The key distinction is the *posterior independence assumption*: the EBNM approach assumes $q(\beta) = \prod_k q(\beta_k)$, so that $E_q[\beta_k\beta_{k'}] = E_q[\beta_k]E_q[\beta_{k'}]$ for $k \neq k'$. This enables a coordinate-ascent EBNM update consistent with the L and F updates. The WLS derivation has been **removed** to avoid confusion; only the EBNM approach is retained below.

6A Setting Up the Objective

First, note that β appears only in the likelihood for the survival data, so we focus on that term. The objective function for q_β is:

$$F\left(q_\beta, g_\beta\right) = E_{q_l,q_\beta}\left[\log P\left(t,\delta|L,\beta\right)\right] + \sum_k E_{q_{\beta_k}}\left[\log\frac{g_{\beta_k}\left(\beta_k\right)}{q_{\beta_k}\left(\beta_k\right)}\right] \tag{47}$$

Using the second-order Taylor approximation from Section 3, we substitute the working response z_i and weights $\mathbf{W}_{ii}$:

$$F(q_\beta, g_\beta) \approx E_{q_l,q_\beta}\left[-\frac{1}{2}\sum_i \mathbf{W}_{ii}\left(z_i - \sum_k l_{ik}\beta_k\right)^2\right] + \sum_k E_{q_{\beta_k}}\left[\log\frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)}\right] \tag{48}$$

We also note an expectation-variance equivalence as follows:

$$E_{q_l}\left[l_{ik}^2\right] = \mathrm{Var}_{q_l}\left[l_{ik}\right] + \left(E_{q_l}\left[l_{ik}\right]\right)^2 \quad \Rightarrow \quad \overline{l_{ik}^2} = \mathrm{Var}_{q_l}\left[l_{ik}\right] + \left(\overline{l_{ik}}\right)^2 \tag{49}$$

6B Coordinate-Ascent Update for β_k

By applying the mean-field independence assumption $q(\beta) = \prod_k q(\beta_k)$, we update a single coefficient β_k by fixing all $\beta_{k'}$ for $k' \neq k$. The survival partial working response (consistent with Eq. 20) is:

$$z_i^{-k} = z_i - \sum_{k' \neq k} l_{ik'} \beta_{k'} \quad \Rightarrow \quad z_i \approx z_i^{-k} + l_{ik}\beta_k \tag{50}$$

Substituting this into equation 48:

$$-\frac{1}{2}\sum_i \mathbf{W}_{ii}(z_i^{-k} - l_{ik}\beta_k)^2 = -\frac{1}{2}\sum_i \mathbf{W}_{ii}\left(\cancel{(z_i^{-k})^2} - 2z_i^{-k}\underline{l_{ik}}\beta_k + \underline{l_{ik}^2}\beta_k^2\right) \tag{51}$$

Taking the expectation with respect to q_l, using $E_{q_l}[l_{ik}] = \overline{l_{ik}}$ and $E_{q_l}[l_{ik}^2] = \overline{l_{ik}^2}$:

$$F(q_{\beta_k}) = \sum_i \left[\mathbf{W}_{ii} z_i^{-k}\overline{l_{ik}}\beta_k - \frac{1}{2}\mathbf{W}_{ii}\overline{l_{ik}^2}\beta_k^2\right] + E_{q_{\beta_k}}\left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)}\right] + C \tag{52}$$

6C Formulating the EBNM Problem

Correction Note

Item 4: The previous document (Sec. 6A, Eq. 91) included $+1/\sigma^2$ in A_k, explicitly accounting for a fixed prior $g(\beta_k) \sim N(0, \sigma^2)$. However, the EBNM solver empirically fits g_{β_k} directly from the data, so adding $1/\sigma^2$ to A_k accounts for the prior **twice**—once through the EBNM prior estimation and again through the explicit term. The corrected A_k omits the $1/\sigma^2$ term, and x_k is updated accordingly. Note also that $\overline{l_{ik}}$ (posterior mean) appears in B_k, and $\overline{l_{ik}^2}$ (full posterior second moment) appears in A_k, encoding an error-in-variables correction: uncertainty in L inflates the effective noise for β_k.

Grouping equation 52 into the standard quadratic form $-\frac{1}{2}A_k\beta_k^2 + B_k\beta_k$:

$$\mathbf{A_k} = \sum_i \mathbf{W}_{ii}\overline{l_{ik}^2} \tag{53}$$

$$\mathbf{B_k} = \sum_i \mathbf{W}_{ii} z_i^{-k}\overline{l_{ik}} \tag{54}$$

This yields the inputs for the 1D EBNM problem with a single pseudo-observation for factor k:

$$x_k = \frac{B_k}{A_k} = \frac{\sum_i \mathbf{W}_{ii} z_i^{-k}\overline{l_{ik}}}{\sum_i \mathbf{W}_{ii}\overline{l_{ik}^2}} \tag{55}$$

$$s_k = \frac{1}{\sqrt{A_k}} \tag{56}$$

For each factor k, we solve the EBNM problem:

$$(\hat{g}_{\beta_k}, \hat{q}_{\beta_k}) = \text{EBNM}(x_k, s_k) \tag{57}$$

Note: The EBNM solver empirically fits g_{β_k} from the data, so no fixed σ^2 is required.

7 Update for τ

Correction Note

Item 5: This section is unchanged from the previous document. It follows Section A.1 of the EBMF paper (Equations A.5–A.11) exactly and has been confirmed correct.

We derive updates to optimize the objective function F over the precision parameters τ identically to the procedure defined in section $A.1$ of the EBMF paper (equations $(A.5) - (A.11)$). This update specifically focuses on the components of F that depend on τ such that

$$F(\tau) = E_{q_l} E_{q_f} \sum_{ij} \frac{1}{2} \log(\tau_j) - \frac{1}{2} \tau_j \left(Y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 + C \tag{58}$$

$$= \frac{1}{2} \sum_{ij} \left[\log(\tau_j) - \tau_j \overline{R^2}_{ij} \right] + C \tag{59}$$

where $\overline{R^2}_{ij}$ is defined by

$$\overline{R^2}_{ij} :- E_{q_l, q_f} \left[\left(Y_{ij} - \sum_{k=1}^{K} l_{ik} f_{jk} \right)^2 \right] \tag{60}$$

$$= \left(Y_{ij} - \sum_k \bar{l}_{ik} \bar{f}_{jk} \right)^2 - \sum_k \left(\bar{l}_{ik} \right)^2 \left(\bar{f}_{jk} \right)^2 + \sum_k \overline{l^2}_{ik} \overline{f^2}_{jk} \tag{61}$$

If $\tau \in \mathcal{T}$ is constrained, we have

$$\hat{\tau} = \arg\max_{\tau \in \mathcal{T}} \sum_{ij} \left[\log(\tau_j) - \tau_j \overline{R^2}_{ij} \right] \tag{62}$$

Assuming constant precision $\tau_{ij} = \tau$ yields

$$\hat{\tau} = \frac{NP}{\sum_{ij} \overline{R^2}_{ij}} \tag{63}$$

and assuming column-specific precisions ($\tau_{ij} = \tau_j$) yields

$$\hat{\tau}_j = \frac{N}{\sum_i \overline{R^2}_{ij}} \tag{64}$$

In equations 63 and 64, the variables N and P refer to the following:

- N: The number of samples/rows in the data matrix Y
- P: The number of features/columns in the data matrix Y

Therefore, for equation 63, NP refers to the total number of entries in the $n \times p$ matrix. This equation sets the precision to the inverse of the average expected squared residual across all N rows and P columns. Furthermore, for equation 64, N, the number of rows, is used because the precision for a specific column is estimated based on the N observations within that column.

8 Algorithm

The following algorithm presents the complete coordinate-ascent variational inference (CAVI) procedure for the supervised matrix factorization model. The algorithm iterates over blocks for L, F, β, and τ. Equation references are to the numbered expressions in the sections above.

Algorithm 1. Supervised MF CAVI Update Procedure

Require: Genomics matrix $Y_{n \times p}$, survival data $(t_i, \delta_i)_{i=1}^n$, number of factors K, convergence tolerance ε.

Initialize ($b = 0$)**:**

- $\overline{L}^{(0)}$, $\overline{F}^{(0)}$ from rank-K SVD of Y; set $\overline{l_{ik}^2}^{(0)} \leftarrow \overline{l_{ik}}^{(0)^2}$, $\overline{f_{jk}^2}^{(0)} \leftarrow \overline{f_{jk}}^{(0)^2}$
- $\overline{\beta_k}^{(0)}$ from Cox PH on initial loadings $\overline{L}^{(0)}$; set $\overline{\beta_k^2}^{(0)} \leftarrow \overline{\beta_k}^{(0)^2}$
- $\hat{\tau}_j^{(0)}$ from column-wise sample variance of Y

For $b = 1, 2, \ldots$ until convergence:

Step 1 — Compute Survival Working Quantities *(once per outer iteration, using posterior means from $b-1$)*

1.1 Compute the linear predictor for each sample i:

$$\hat{\eta}_i = \sum_{k=1}^{K} \overline{l_{ik}}\,\overline{\beta_k}$$

1.2 Evaluate Cox score u_i and negative Hessian W_{ii} at $\hat{\boldsymbol{\eta}}$:

$$u_i = \left.\frac{\partial \ell_{\text{Cox}}}{\partial \eta_i}\right|_{\hat{\eta}_i}, \qquad \mathbf{W}_{ii} = -\left.\frac{\partial^2 \ell_{\text{Cox}}}{\partial \eta_i^2}\right|_{\hat{\eta}_i} \geq 0$$

1.3 Compute the working response (Sec. 3):

$$z_i = \hat{\eta}_i + \frac{u_i}{\mathbf{W}_{ii}}$$

Step 2 — Update Factors for $k = 1, \ldots, K$:

(a) Update $q(l_k)$, $g(l_k)$ — Patient Loadings (Eqs. 30–33)

Define partial residuals with raw components (Eqs. 19, 20); evaluate at posterior means to obtain $\overline{R_{ij}}^{-k}$ and $\overline{z_i}^{-k}$:

$$R_{ij}^{-k} = y_{ij} - \sum_{k' \neq k} l_{ik'} f_{jk'}, \quad \overline{R_{ij}}^{-k} = y_{ij} - \sum_{k' \neq k} \overline{l_{ik'}}\,\overline{f_{jk'}}$$

$$z_i^{-k} = z_i - \sum_{k' \neq k} l_{ik'} \beta_{k'}, \quad \overline{z_i}^{-k} = z_i - \sum_{k' \neq k} \overline{l_{ik'}}\,\overline{\beta_{k'}}$$

Compute EBNM coefficients (Eqs. 30–31), solve, and store:

$$\mathbf{A_{ik}} = \sum_j \hat{\tau}_j\,\overline{f_{jk}^2} + \mathbf{W}_{ii}\,\overline{\beta_k^2}, \qquad \mathbf{B_{ik}} = \sum_j \hat{\tau}_j\,\overline{R_{ij}}^{-k}\,\overline{f_{jk}} + \mathbf{W}_{ii}\,\overline{z_i}^{-k}\,\overline{\beta_k}$$

$$(\hat{g}_{l_k}, \hat{q}_{l_k}) = \text{EBNM}\Big(x_i = \tfrac{B_{ik}}{A_{ik}},\; s_i = \tfrac{1}{\sqrt{A_{ik}}}\Big) \Rightarrow \text{store } \overline{l_{ik}},\ \overline{l_{ik}^2}$$

(b) Update $q(f_k)$, $g(f_k)$ — Biological Factors (Eqs. 43–46)

Recompute $\overline{R_{ij}}^{-k}$ using updated $\overline{l_{ik}}$. Compute EBNM coefficients (Eqs. 43–44), solve, and store:

$$\mathbf{A_{jk}} = \sum_i \hat{\tau}_j\,\overline{l_{ik}^2}, \qquad \mathbf{B_{jk}} = \sum_i \hat{\tau}_j\,\overline{R_{ij}}^{-k}\,\overline{l_{ik}}$$

$$(\hat{g}_{f_k}, \hat{q}_{f_k}) = \text{EBNM}\Big(x_j = \tfrac{B_{jk}}{A_{jk}},\; s_j = \tfrac{1}{\sqrt{A_{jk}}}\Big) \Rightarrow \text{store } \overline{f_{jk}},\ \overline{f_{jk}^2}$$

(c) Update $q(\beta_k)$, $g(\beta_k)$ — Survival Coefficients (Eqs. 53–56)

Compute updated survival partial working response using current $\overline{l_{ik}}$, $\overline{\beta_k}$ (raw definition evaluated at posterior means yields $\overline{z}_i^{-k}$):

$$z_i^{-k} = z_i - \sum_{k' \neq k} l_{ik'}\,\beta_{k'}, \qquad \overline{z}_i^{-k} = z_i - \sum_{k' \neq k} \overline{l_{ik'}}\,\overline{\beta_{k'}}$$

Compute EBNM coefficients (*no* $1/\sigma^2$ *term in* A_k; Eqs. 53–54), solve, and store:

$$\mathbf{A_k} = \sum_i \mathbf{W}_{ii}\,\overline{l_{ik}^2}, \qquad \mathbf{B_k} = \sum_i \mathbf{W}_{ii}\,\overline{z}_i^{-k}\,\overline{l_{ik}}$$

$$(\hat{g}_{\beta_k}, \hat{q}_{\beta_k}) = \mathrm{EBNM}\Big(x_k = \tfrac{B_k}{A_k},\ s_k = \tfrac{1}{\sqrt{A_k}}\Big) \ \Rightarrow\ \text{store } \overline{\beta_k},\ \overline{\beta_k^2}$$

end for (over $k = 1, \dots, K$)

Step 3 — **Update $\hat{\tau}$** (Eq. 64)

3.1 Compute expected squared residual for each entry (i, j):

$$\overline{R^2}_{ij} = \Big(Y_{ij} - \sum_k \overline{l_{ik}}\,\overline{f_{jk}}\Big)^2 + \sum_k \Big[\overline{l_{ik}^2}\,\overline{f_{jk}^2} - \overline{l_{ik}}^2\,\overline{f_{jk}}^2\Big]$$

3.2 Update column-specific precision:

$$\hat{\tau}_j = \frac{N}{\sum_i \overline{R^2}_{ij}}$$

Step 4 — **Check Convergence**

Compute maximum absolute change in loadings and survival coefficients:

$$\Delta_L = \max_{i,k} \left| \overline{l_{ik}}^{\,\text{new}} - \overline{l_{ik}}^{\,\text{old}} \right|, \qquad \Delta_\beta = \max_k \left| \overline{\beta_k}^{\,\text{new}} - \overline{\beta_k}^{\,\text{old}} \right|$$

If $\max(\Delta_L, \Delta_\beta) < \varepsilon$: stop. **Else**: continue to $b + 1$.

Return: posterior means $\overline{L}$, $\overline{F}$, $\overline{\boldsymbol{\beta}}$; second moments $\overline{L^2}$, $\overline{F^2}$, $\overline{\boldsymbol{\beta}^2}$; noise precision $\hat{\tau}$